

PUBLIC SUBMISSION

As of: March 21, 2025
Received: March 14, 2025
Status: [REDACTED]
Tracking No. m89-a631-55li
Comments Due: March 15, 2025
Submission Type: API

Docket: NSF_FRDOC_0001

Recently Posted NSF Rules and Notices.

Comment On: NSF_FRDOC_0001-3479

Request for Information: Development of an Artificial Intelligence Action Plan

Document: NSF_FRDOC_0001-DRAFT-1473

Comment on FR Doc # 2025-02305

Submitter Information

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Roschelle



General Comment

The January 23 Executive Order (EO) on artificial intelligence (AI) states: "It is the policy of the United States to sustain and enhance America's global AI dominance in order to promote human flourishing, economic competitiveness, and national security."

This response recommends:

1. NSF's AI Action Plan should aim to achieve the national policy goal of human flourishing.
2. NSF's AI Action Plan should include education in its approach to addressing this national policy goal.
3. NSF's AI Action Plan should enumerate specific steps NSF will take to achieve the national policy goal through its existing educational mission, such as investigating how STEM educational pathways can leverage AI; advancing measurement of factors closely; and increasing each American's opportunity to learn about AI.

Details in support of these recommendations follow in the attachment, organized by three questions: why, how, and what.

Attachments

Comment on NSF_FRDOC_0001-3479 Roschelle

Comment on NSF AI RFI

Comment on NSF_FRDOC_0001-3479

Jeremy Roschelle, Bill Penuel and Jeanne Century

Submitted March 14, 2025

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Details in support of these recommendations follow, organized by three questions: why, how, and what.

WHY INCLUDE "HUMAN FLOURISHING" in NSF's AI ACTION PLAN?

NSF is uniquely positioned to support human flourishing. Although NSF plays an important role in supporting the other goals in the EO—national security and economic competitiveness—other agencies have clear missions to advance these areas, too. NSF, however, has a clear opportunity to lead in human flourishing in ways that other agencies do not.

First, NSF's leadership in science, mathematics, engineering and technology (STEM) has already expanded the well-being of all Americans. For example, NSF's investments in the underlying technology of the Internet have transformed communication, and interdisciplinary advances in computing and physics led to MRI technology that has revolutionized medicine. In education, applications of AI to support intelligent tutoring, for example, set the stage for contemporary advances in AI-based tutors now being delivered by technology companies such as Duolingo, Khan Academy, Carnegie Learning, and more. NSF has an unparalleled track record of linking scientific discovery to human flourishing already, and can do more to achieve human flourishing in an age of AI.

Second, improving education is essential to human flourishing. Americans will need more and better ways to learn about AI to enable each person to realize the benefit of AI in their life.

NSF-funded researchers already pursue evidence-based approaches to teaching and learning, and these approaches move from foundational knowledge to active use in curriculum resources, teacher training, educational technologies, assessment techniques, and more. NSF's education mission in STEM already includes AI in its scope, and within this existing scope, there is more NSF can do to produce the knowledge base needed so that each American can flourish by better incorporating AI productively into their learning and their long-term pursuits.

Third, the review criteria that NSF applies in evaluating every grant proposal already specify a clear place to advance the "human flourishing" policy goal. One of two review criteria is "Broader Impacts." When they write grant proposals, researchers can readily incorporate specific goals related to human flourishing with AI as Broader Impacts and state how they will measure progress towards these goals. Reviewers can recommend grant proposals on the strength of their contribution to knowledge about human flourishing with AI. Thus, NSF can readily incorporate the policy goal of human flourishing into its core work of inviting and reviewing proposals.

To summarize why NSF's AI Action Plan should address the national policy goal of increasing human flourishing: NSF already has a track record that links its scientific discoveries to human flourishing, has the educational mission essential to human flourishing with AI, and has a clear place to address human flourishing in the heart of its grant review process.

HOW NSF CAN ADVANCE THE POLICY GOAL BY FOCUSING ON EDUCATION

Human flourishing is a future-oriented concept, and **today's learners are the future.**

The field of Education has been a strong early adopter of generative AI and the ways AI can directly contribute to flourishing of learners, and NSF's past investments have been critical in showing how it can. Examples abound: AI can help improve educators' productivity; AI can strengthen students' learning experiences; AI can enable students to ask stronger scientific questions and develop stronger scientific arguments; AI can help students learn to solve problems together; and AI can support tutoring in STEM.

Conversely, education has also been a primary arena for identifying the risks AI can pose to human flourishing. For example, if students use AI to do assignments in place of their own thinking, they will not learn the lesson content and in turn, will not be as likely to flourish. Similarly, if they use AI as a crutch in challenging learning situations, their reasoning may atrophy. We are just at the beginning of developing knowledge about human flourishing with AI in education, and a strong program of NSF research is critical to ensure the strengths of AI are well-leveraged and risks are mitigated.

Education is the societal activity that brings most Americans into contact with AI, and thus presents an unmatched opportunity for NSF researchers to gather evidence and build knowledge

about human flourishing with AI. **Thus, including education in NSF's AI Action Plan will strongly advance the national "human flourishing" policy goal.**

WHAT NSF CAN DO IN STEM EDUCATION RESEARCH TO ACHIEVE THE POLICY GOAL

First, NSF can lead through research on how educational pathways can leverage AI for student success.

NSF-supported research can generate knowledge about how educators and learners use AI to develop their knowledge and skills over time; how AI can be part of well-designed, efficacious learning experiences; and how both can lead to later successes in education, careers, and playing productive roles in society. Specifically, NSF has a long history of recognizing that successful education cannot only address an isolated unit of study, test outcome, or grade level, but rather must include a range of pathways through which individual learners advance their long-term goals. Without question, all learner pathways of the future will come to include AI. By continuing its pathway focus as AI becomes commonplace in education, NSF can be a leader in generating knowledge about the role AI can play in realizing the potential of every American while strengthening their well-being.

Second, NSF can support the development of measures to track progress towards human flourishing with AI in education.

To achieve the policy goal, it is essential to have the necessary tools to demonstrate progress to the American people—and so that innovators can benchmark their progress. NSF-funded research has already made enormous headway in developing measures of key factors related to human flourishing. A partial list includes:

- **STEM knowledge** (improvement in the skills, understanding, and dispositions that lead to further participation in STEM learning and careers);
- **metacognition** (the ability for each student to take charge of their own learning);
- **agency** (the ability for each student to make choices that align to and advance their own learning goals);
- **creativity and innovation** (the ability for each student to advance their own ability to create new things purposefully); and
- **teamwork** (the ability for each student to work with others to advance knowledge and reasoning).

For example, the NSF's AI Institutes (including five Institutes that focus on education) are already making significant advances in how to measure these outcomes efficiently and effectively. **By continuing to support the development of measures, NSF can translate the broad policy goal of "human flourishing" into concrete, measurable objectives that leaders across the United States can more readily apply and advance.**

Third, NSF can support research on innovations that enable people across the lifespan and in varied settings to develop knowledge and skills necessary for their mastery of AI.

To flourish, learners of every age need more and better understanding of AI. NSF is already a leader in supporting efforts to effectively integrate technological advances (e.g., computer science, cybersecurity, data science) into education at every level. NSF's educational research investments have created outstanding opportunities for students to develop technological skills and knowledge starting in childhood, through adolescence, and as they enter adulthood and the workforce. Now, it is essential that NSF invest in research efforts that build knowledge essential to enabling **every person in the United States to learn AI in ways that fit their settings, needs, and strengths.**

CONCLUSION

NSF is well-positioned to lead the United States on the national policy goal of "human flourishing" with AI. By including its historic role in education within the scope of its AI Action Plan, NSF can lead the nation in producing benefits directly for each American and thus ensure the AI Action benefits each citizen. By building on the STEM education research already completed about how people learn, NSF's action plan can have specific steps for further research in education that ensure the AI Action Plan will contribute to human flourishing.

Therefore, we recommend:

1. NSF's AI Action Plan should aim to achieve the national policy goal of human flourishing.
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